António José Freitas Machado Dórdio Gomes

MECHANICAL ENGINEER
PRODUCT DEVELOPMENT & COMPOSITE STRUCTURES

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Portfolio Website

Profile

Mechanical engineer specialised in product development and mechanical design, with experience in CAD modelling, prototyping, structural analysis and component validation through Formula Student and aerospace engineering projects. Further details and selected engineering projects can be found in the **engineering portfolio**.

Professional Experience

Mechanical Design Engineer – UAVision — Jun 2026 – Present

- Integrate electronics into **Veteras** (all-electric heavy-lift VTOL, up to **200 kg payload**), **MPAV-S** (multipurpose modular UGV for logistics and casualty evacuation) and **UCDL Z10** (modular, gyrostabilised anti-UAS jammer)
- Design sheet-metal parts in **Fusion 360**; **operate laser cutting and press brake machines** for in-house production
- **Redesigned and manufactured UCDL Z10 machined components as sheet-metal parts in-house**, reducing **total jammer assembly weight by 23%**
- Performed **Ansys static and modal analyses** of a helicopter battery and gimbal support: **six directional load cases** per **ASTM F3842-26**; assessed **resonance risk** against experimentally validated helicopter excitation frequencies
- Contributed to CAD design, manufacturing, assembly and integration troubleshooting for the **ELANUS launcher**; **ELANUS** was one of **five finalists from 140+ European entries** in the EDA Sentinel Strike Challenge
- **FDM-print parts for electronics integration** with application-specific filament selection; **develop prototypes and improve existing products**, primarily **Veteras and UCDL Z10**

Graduate Researcher — Master's Thesis — Sep 2025 – Mar 2026

Department of Aerospace Science and Technology — Politecnico di Milano, Italy

- Developed **Abaqus/Explicit FE models** for **low-velocity impact damage** in **Glass/PPS laminates**
- Implemented **Hashin intralaminar damage** and **cohesive zone modelling** for fibre fracture and delamination
- Introduced a **bi-phasic shell–solid discretisation** to capture **indentation** and **matrix-dominated deformation**
- Validated simulations with **impact tests** and **ultrasonic C-scan data**, predicting **delamination area within ∼15%**

Internship — WEG — Bachelor's Thesis — Feb 2023 – Jun 2023

Product Development and Certification Department

- Developed **ANSYS thermal models** to evaluate **heat dissipation** in electric motor housings
- Quantified **cooling performance differences** between **steel and cast iron motor housings** (up to **∼52%**)
- Performed **parametric analysis of fin geometry**, recommending **chamfered fins** for steel housings
- Validated simulations with **experimental tests**, achieving **7.7% deviation**

Engineering Experience

Formula Student FEUP — Electric Formula Racing Prototype Development

Wishbones & Vibrations Lead — **Sep 2024 – Aug 2025**

- **Led a 2-person sub-team** developing and manufacturing suspension wishbones using **filament-wound CFRP tubes adhesively bonded to 7075-T6 aluminium connectors**
- Performed **Abaqus simulations** of aluminium connectors across **25 suspension load cases** and **adhesive joints** validated through correlation with **tensile tests**
- Manufactured and tested **∼40 representative tube–connector joint specimens** in **tensile tests**
- Investigated **adhesive joint failures** and implemented a **bi-adhesive bonding solution** validated up to $80\,^{\circ}\mathrm{C}$
- Supported **prototype development** with **CAD, CNC parts, adhesive bonding and 3D-printed components**

Transmission Engineer — **Sep 2023 – Sep 2024**

- Designed **in-house drivetrain components** (**tripod housing, driveshaft and halfshafts**), in Inventor, for the car
- Performed **structural and safe-life fatigue analysis** of rotating drivetrain components
- Verified **spline interfaces** and **standards compliance** using **involute spline inspection**
- Contributed to a **4 kg drivetrain subsystem mass reduction** vs previous prototype

Education

MSc in Mechanical Engineering – Aeronautical Structures and Vehicles — 2023 – 2026

Faculty of Engineering, University of Porto (FEUP)

Mechanical Engineering Bachelor — 2020 – 2023

Faculty of Engineering, University of Porto (FEUP)

Awards

- **2025 (EV Class) — Formula Student Germany**: 3rd Cost · 9th Business · 25th Design · 36th Overall — **Presented Cost & Cost Final. Formula Student Portugal**: 1st Business · 2nd Design · 3rd Cost · 4th Overall — **Presented Cost & Cost Final. Formula Student Netherlands**: 1st Business · 3rd Cost.
- **2024 (Concept Class) — Formula Student Switzerland**: 1st Overall · 1st Design · 1st Cost · 1st Business. **Formula Student UK**: 2nd Overall · 1st Cost · 2nd Lap Time · 3rd Design — **Presented Design & Cost Final. Formula Student Portugal**: 1st Overall · 1st Design · 1st Cost · 1st Business — **Presented Cost.**

Academic & Volunteering Activities

NEEM FEUP - Pedagogic Department — Mar 2021 – Jun 2025

Mechanical Engineering Students' Association — FEUP

- Organized MecGames (Engineering Competition) and WelcoMec (First-Year Mentoring Program)
- Staff member for Connecting Gears 2022–25 (Mechanical Engineering Technical Conference)
- Staff member for ENEEM 2023 (National Meeting of Mechanical Engineering Students)

Volunteer Work – Banco Alimentar Contra a Fome do Porto — **May 2023**

Certifications

CSWA — Certified SolidWorks Associate, Dassault Systèmes, Aug 2023

Composites Design Workshop XXVII, Prof. Steve Tsai, Stanford University, Jul 2024

Technical Skills

- Abaqus
- Fusion 360
- ANSYS
- Python
- MATLAB
- Excel
- Inventor
- SolidWorks
- LaTeX

Languages

Portuguese	Native
English	C1
Spanish	B1
French	A2

Engineering Focus

Product Development
Mechanical Design
CAD Modelling
Prototyping
Structural Analysis
Component Validation
Composite Structures
Additive Manufacturing
Finite Element Modelling